

Translated

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Flows with demands

In a normal flow network the flow of an edge is only limited by the capacity $c(e)$ from above and by 0 from below. In this article we will discuss flow networks, where we additionally require the flow of each edge to have a certain amount, i.e. we bound the flow from below by a **demand** function $d(e)$:

$$d(e) \leq f(e) \leq c(e)$$

So next each edge has a minimal flow value, that we have to pass along the edge.

This is a generalization of the normal flow problem, since setting $d(e) = 0$ for all edges e gives a normal flow network. Notice, that in the normal flow network it is extremely trivial to find a valid flow, just setting $f(e) = 0$ is already a valid one. However if the flow of each edge has to satisfy a demand, than suddenly finding a valid flow is already pretty complicated.

We will consider two problems:

1. finding an arbitrary flow that satisfies all constraints
2. finding a minimal flow that satisfies all constraints

Finding an arbitrary flow

We make the following changes in the network. We add a new source s' and a new sink t' , a new edge from the source s' to every other vertex, a new edge for every vertex to the sink t' , and one edge from t to s . Additionally we define the new capacity function c' as:

- $c'((s', v)) = \sum_{u \in V} d((u, v))$ for each edge (s', v) .
- $c'((v, t')) = \sum_{w \in V} d((v, w))$ for each edge (v, t') .
- $c'((u, v)) = c((u, v)) - d((u, v))$ for each edge (u, v) in the old network.
- $c'((t, s)) = \infty$

If the new network has a saturating flow (a flow where each edge outgoing from s' is completely filled, which is equivalent to every edge incoming to t' is completely filled), then the network with demands has a valid flow, and the actual flow can be easily reconstructed from the new network. Otherwise there doesn't exist a flow that satisfies all conditions. Since a